
LEARNING TO WALK RANDOMLY

ABSTRACT

Bipedal locomotion in continuous-control environments is difficult due to high-dimensional actions and sensitivity to exploration strategies. We apply Truncated Quantile Critics (TQC), an entropy-regularised distributional actor-critic method, to the rldurham/Walker Box2D environment in easy and hardcore settings. Our agent uses five quantile critics, observation normalisation, and critic LayerNorm. It learns to walk in both settings, achieving best episode returns of 244.2 on easy and 224.6 on hardcore. However, late-stage instability persists in both settings.

1 METHODOLOGY

1.1 TASK AND ALGORITHM

The rldurham/Walker environment is a Box2D continuous-control task [10] in which a bipedal agent must learn to walk efficiently across either the easy or hardcore setting. The easy environment consists of slightly uneven terrain, whereas the hardcore setting has randomly generated ladders, stumps and pitfalls. The agent observes 24 continuous dimensions and outputs 4 continuous actions bounded to $[-1, 1]$, controlling torque at each hip and knee joint. The goal is to achieve a competitive score within 1000 episodes (easy) or 2000 episodes (hardcore).

	Name	Min	Max
0	Hip 1 (Torque/velocity)	-1	+1
1	Knee 1 (Torque/velocity)	-1	+1
2	Hip 2 (Torque/velocity)	-1	+1
3	Knee 2 (Torque/velocity)	-1	+1

Table 1: The continuous action space is between -1 and $+1$ so `torch.tanh` is sensible in the last actor layer.

We use Truncated Quantile Critics (TQC), an off-policy distributional actor-critic method designed to reduce Q-value overestimation [6]. Rather than predicting a single scalar return, each critic predicts a set of quantiles that approximate the return distribution. In our implementation, each of the $N=5$ critics outputs $M=25$ quantiles, which are trained with the quantile Huber loss [2]:

$$\mathcal{L}_{\text{critic}} = \frac{1}{|B|} \sum_B \sum_{i=1}^M \frac{1}{K} \sum_{j=1}^K |\tau_i - \mathbf{1}(\delta_{ij} < 0)| \cdot H(\delta_{ij}) \quad (1)$$

where $\delta_{ij} = y_j - z_i$ is the TD error between predicted and target quantiles, and $\tau_i = \frac{i-0.5}{M}$ is the quantile midpoint. Quantile predictions from all N target critics are pooled, sorted, and the top $d=2$ per critic dropped, removing the 10 most optimistic of 125 total quantiles:

$$y = r + \gamma(1 - \mathbb{K}_{\text{term}})(z_{\text{trunc}} - \alpha \log \pi(a' | s')) \quad (2)$$

where $\mathbb{K}_{\text{term}}$ is 1 for terminal transitions and 0 otherwise. The policy follows Soft actor critic (SAC), a stochastic Gaussian actor with tanh squashing, trained to maximise expected Q-value while maintaining entropy[4]:

$$\mathcal{L}_{\text{actor}} = \mathbb{E} \left[\alpha \log \pi(a | s) - \frac{1}{NM} \sum_{n,m} z_{nm}(s, a) \right] \quad (3)$$

The entropy temperature α is automatically tuned to $-\dim(A)$, and target networks are updated via Polyak averaging ($\tau = 0.005$) [7]. TQC is advantageous because its distributional critics provide stable value estimates as Q-values shift from -100 to 240+, and its stochastic policy offers natural exploration for discovering coordinated gait patterns.

1.2 AGENT ARCHITECTURE

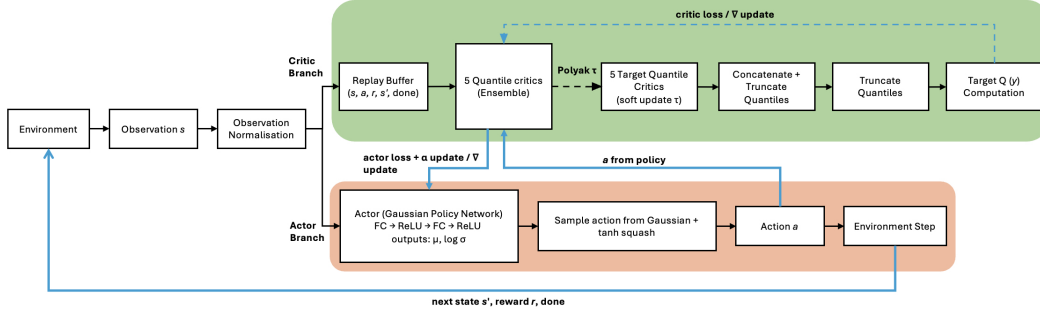


Figure 1: Diagram of TQC architecture. Critic branch shown in green. Actor branch shown in orange. Dashed arrows represent gradient updates.

Our agent architecture is shown in Figure 1. The actor is a stochastic Gaussian network with two hidden layers and ReLU activations, outputting a mean and log standard deviation for each action dimension (clamped to $[-5, 2]$ for stability). Actions are squashed through tanh to enforce $[-1, 1]$ bounds, with the standard log-probability correction applied [4]. Five quantile critics each take the concatenated state-action pair through two hidden layers with LayerNorm and ReLU, outputting 25 quantile values. LayerNorm stabilises internal activations as Q-value estimates shift from -100 to 240+ during training [1]. The actor does not use LayerNorm. Each critic has a corresponding target network, initialised as an identical copy.

All observations pass through a running normaliser that tracks online mean and variance. This rescales inputs to approximately zero mean and unit variance. This produced one of the clearest performance gains in our ablations, as the 24-dimensional observation space contains features on different scales. The lidar readings are bounded in $[0, 1]$, while angular velocities can be much larger. Transitions are stored in a replay buffer of capacity 1M. Both environments use the same architecture, differing only in hidden layer width (256 for easy, 400 for hardcore) and warmup period (5,000 vs 10,000 random steps), giving the hardcore agent additional representational and more diverse initial transitions for the harder terrain.

	Easy	Hardcore
Hidden layers width	256	400
Warmup steps	5,000	10,000
Target entropy	-4 (fixed)	-4 $\rightarrow$ -7 (annealed)
Actor objective	Mean	Median
Max episodes	1,000	2,000
Reward scaling	1.0	5.0

Table 2: Key differences between easy and hardcore configurations. Other hyperparameters are shared: batch size=256, LR=3e-4, $\gamma = 0.99$, $\tau = 0.005$, gradient clip=1.0, updates per step=2.

1.3 TRAINING PROCEDURE

Training follows an episodic structure where the agent collects experience then performs 2 gradient updates per step at episode end [8]. The first 5,000 steps (10,000 for hardcore) use uniform random actions. The actor and α update every second critic update (policy delay 2) [3], and gradient norms are clipped to 1.0 [9]. For hardcore, three modifications were made: target entropy is annealed from -4 to -7 over 800,000 steps to reduce late-stage instability, the actor uses median Q-value across critics for robustness to outlier estimates, and only the terminated signal is used as done, preventing the agent from treating the time limit as failure. Additionally, rewards are scaled by a factor of 5.0 in the hardcore environment to amplify the learning signal relative to the entropy regularisation term, which otherwise dominates when raw rewards are small during early training.

2 CONVERGENCE RESULTS

2.1 EASY ENVIRONMENT

We conducted 14+ ablations in the easy environment, varying hidden layer width, truncation level, gradient clipping, updates per step, and actor objective. These changes did not improve beyond the existing ~ 238 plateau. However, implementing observation normalisation and critic LayerNorm increased the convergence level, at the cost of slightly delayed convergence. Three further modifications were tested, beginning with prioritised experience replay (PER). PER reached a score of 200 by episode 100, whereas the non-PER configuration first reached 200 at episode 136, but still plateaued at ~ 237 . This suggests limited benefit once the distributional critics were already learning effectively. Another modification was entropy annealing, which reduces exploration over episodes. This worsened late-stage collapses, with the minimum episode return falling from -73 under fixed entropy to -193 under annealing.

The final configuration for the easy environment combined observation normalisation and critic LayerNorm with refined hyperparameters (Table 2). This raised the best score from 238 to 244.2. As shown in Figure 2, the agent first exceeds 200 at episode 136. The rolling mean surpasses 240 by episode 386. Episodes 700-900 were remarkably stable, with a mean of 243.1 and a standard deviation of 0.4. There were moments of late-stage instability in episodes 900-1000, where the mean dipped to 236.9. This is consistent with the fixed entropy term maintaining residual exploration even after a strong gait had been learned. The agent sustained performance around 243, with a peak of 244.2. Overall, the easy agent shows clear convergence by around episode 386 with only occasional late-stage regressions.

2.2 HARD ENVIRONMENT

For the hardcore environment, we conducted 10 experiments over 2000 episodes. The main ablation finding was that the easy configuration did not transfer directly. We found that hidden width 256 with 5,000 warmup steps performed poorly, whereas increasing capacity to 400 hidden units and extending warmup to 10,000 steps enabled learning to ~ 207 , but with high variance. A split-lidar architecture was also ineffective, never exceeding -60 after 800 episodes. Entropy annealing from -4 to -7 reached approximately 201 with 3 updates per step, but remained unstable. Reverting to 2 updates per step improved stability slightly, to approximately 202. A repeated run with the same configuration also failed to learn, indicating substantial sensitivity to initialisation. The final configuration combined the larger network and longer warmup with entropy annealing, median critic aggregation, true-termination handling, and a truncation level of $d = 2$, achieving a best score of 224.6.

As shown in Figure 3, the final hardcore agent first exceeded 200 at episode 263, with the rolling mean crossing 150 by episode 841 before plateauing at approximately 180 from episode 1400 onwards. In our experiments, scores above 200 corresponded to competent navigation, making 224.6 a strong result in this environment. However, robustness remained incomplete. Although 82% of the final 400 episodes exceeded 200, 36 episodes in the same window still fell below zero, and the standard deviation remained above 80. Taken together,

these results indicate partial convergence in the hardcore environment: performance improves steadily and reaches a competent regime, but continued high variance shows that full convergence was not achieved within 2000 episodes.

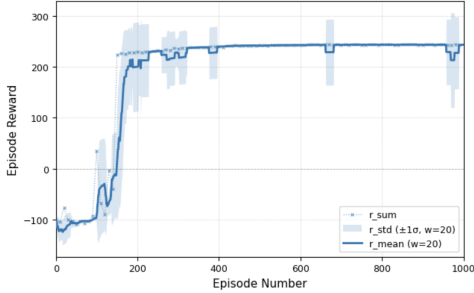


Figure 2: Agent Learning curve for Easy environment. Agent converges around episode 386.

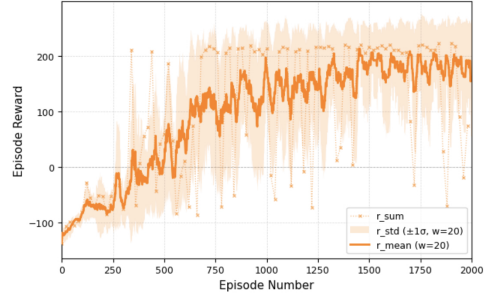


Figure 3: Agent Learning curve for hardcore environment. Agent improves over 2000 episodes, no strong convergence indications.

3 LIMITATIONS

In the easy environment, the main remaining limitation is late-stage instability after entering a strong convergence phase. Although the policy achieved near-ceiling performance, regressions around episodes 700 and 950 in Figure 2 indicate that the final behaviour was not fully robust. One plausible explanation is that the fixed entropy term maintained residual exploration even after a strong gait had been learned. Entropy annealing was tested to address this in Section 2.1, but in the easy environment it worsened collapse severity rather than improving stability. Additionally, the running observation statistics continued to shift late in training, which may also have contributed to the residual variance.

In the hardcore environment, the main limitation is incomplete robustness across procedurally generated terrain. As shown in Figure 3, although 82% of the final 400 episodes exceeded 200, standard deviation remained above 80 and 36 episodes in the same window still fell below zero. This indicates that the agent handled most terrain layouts competently, but remained vulnerable on particularly challenging configurations. Learning was also substantially slower than in the easy environment, and a repeated run with the same configuration failed to learn, indicating sensitivity to initialisation. As in the easy setting, the running observation statistics continued to shift late in training, and the greater diversity of hardcore states may have amplified this effect, contributing to the persistent variance observed throughout training.

4 FUTURE WORK

For future work in the easy environment, adaptive entropy tuning in SAC [4] could replace fixed entropy regularisation, reducing exploration automatically once the policy stabilises. This would directly target the late-stage instability observed in episodes 900-1000, where residual stochasticity may have prevented full policy commitment. Reward shaping to place greater emphasis on forward velocity could also improve on the current best score of 244.2, as the existing reward appears to favour a conservative but highly stable gait.

For future work in the hardcore environment, extending training beyond 2000 episodes could yield further gains, as the rolling mean in Figure 3 was still improving gradually at termination, suggesting that learning had not fully plateaued. Modifying the observation with short-term history using a Long Short-Term Memory (LSTM) [5] or Transformer encoder [11] could provide temporal context for anticipating upcoming obstacles.

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